Syllabus for Calculus-Based Physics II: Electricity and Magnetism (PHYS180)

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Abstract

The concepts of calculus-based electromagnetism will be presented within the context of interactive problem-solving. The course will begin with the concepts of electric charge, electrostatics, electric potential, and capacitance. Next, current and DC circuits will be covered. Magnetostatics and magnetic induction follow DC circuits, concluding with AC circuits. Finally, seleted topics in electromagnetic waves and optics will be presented. The course work will include interactive computational exercises, analytic textbook problems, group-designed projects, and lab-based activities.

Pre-requisites: *PHYS150 or PHYS135A, MATH142 (may be taken concurrently)*
Course credits, Liberal Arts Categorization: *4 Credits, NSIN, QR2*
Regular course hours and location: *MWF - 13:00-15:00, SLC 228.*
Instructor contact information: *Discord: 918particle, Email: jhanson2@whittier.edu, Office: SLC 222.*
Office hours: *Book online appointments:* `https://calendar.app.google/xjeWMYHSuuWvEYKMA`.
Attendance/Absence: *Students needing to reschedule midterms must notify the professor a few days in advance.*
Late work policy: *Late work is generally not accepted, but is left to the discretion of the instructor.*
Text: *University Physics, vol. 2:* `https://openstax.org/details/books/university-physics-volume-2`.
Homework: *Homework exercises and content will be managed by ExpertTA. This online system costs $35 and will provide hints, feedback, and automatically grade student work. Please click this link to register:* `https://reg.theexpertta.com/USB06CA-080799-4T2`.
Grading: *The course grade will be a weighted average of assignment scores, and the weights are listed in Tab. 1.*
Grade Settings: $\geq 60\%, < 70\% = D$, $\geq 70\%, < 80\% = C$, $\geq 80\%, < 90\% = B$, $\geq 90\%, < 100\% = A$. *Pluses and minuses: 0-3% minus, 3%-6% straight, 6%-10% plus (e.g. 79% = C+, 91% = A-).*
ADA Statement on Disability Services: *Whittier College is committed to make learning experiences as accessible as possible. If you experience physical or academic barriers due to a disability, you are encouraged to contact Student Disability Services (SDS) to discuss options. To learn more about academic accommodations, email disabilityservices@whittier.edu, call (562) 907-4825, or go to SDS which is located on the ground floor of Wardman Library.*
Academic Honesty: `https://www.whittier.edu/policies/academic/honesty`
Course Objectives:

- To solve word problems in physics and mathematics, to construct mathematical models of systems
- To match mathematical models of physical systems to experimental results
- To apply logical thinking to conceptually-posed physics problems
- To practice using scientific instrumentation, to collect and analyze data, and to communicate results
- To practice written and oral expression of scientifically technical ideas

Assignment	Weight	Date
Daily exercises	10 %	Completed during class
Homework sets and labs	30 %	Weekly on Fridays
First Midterm	20 %	March 9th, 2026 (take-home style on Units 0-2)
Second Midterm	20 %	May 4th, 2026 (take-home style on Units 3-5)
Final Project Presentation	20 %	April 27th and 29th, 2026 (in class)

Table 1: Grade weighting. The final project presentation may be given via **Option A** or **Option B** (see below).

Course Outline:

1. **Unit 0:** Review of pre-requisite courses, and electrostatics.
   (a) Unit analysis, kinematics, and Newton's Laws
   (b) Work and energy, momentum
   (c) Electrostatics, I - **Chapters - 5.1-5.7, 6.1-6.4**
      i. The Coulomb force, Newton's Second Law for electric charge
      ii. Electric fields of charge distributions
      iii. Guass' Law for charge distributions
   (d) Electrostatics, II - **Chapters 7.1-7.6**
      i. Electric potential and potential energy
      ii. Electric potential and fields
      iii. Equipotentials and conductors
      iv. Applications of electrostatics
2. **Unit 1:** Capacitors and DC current
   (a) Capacitors and capacitance - **Chapters 8.1-8.4**
      i. Capacitance and capacitors
      ii. Series and parallel capacitors, energy considerations
   (b) DC current - **Chapters 9.1-9.5**
      i. Definition of current, Ohm's law
      ii. Energy and power in DC current
3. **Unit 2:** DC Circuits
   (a) DC circuit analysis - **Chapters 10.1-10.3**
      i. Electromotive force
      ii. Resistors in series and parallel
      iii. Kirchhoff's rules
   (b) Applications - **Chapters 10.4-10.6**
      i. Instruments to measure current and voltage
      ii. RC circuits
      iii. Household wiring
4. **First Midterm** - due March 9th, 2026.
   (a) The midterm will be posted on Moodle on Friday, March 6th.
   (b) The exam will be completed at home, with an open-book/open-note policy. The format includes multiple choice, written exercises, and design problems.
   (c) The exam is due in PDF form on Monday, March 9th, to be submitted via Moodle.
5. **Unit 3:** Magnetism I
   (a) Magnetostatics I - **Chapters 11.1-11.7**
      i. Lorentz force and magnetic field lines
      ii. Motion of free charge in magnetic fields
      iii. Force on current-carrying conductors
      iv. The Hall effect
   (b) Magnetostatics II - **Chapters 12.1-12.7**
      i. Biot-Savart law
      ii. Ampère's Law: magnetic fields created by current
6. **Unit 4:** Magnetism II
   (a) Magnetic induction - **Chapters 13.1-13.7**
      i. Faraday and Lenz's laws
      ii. Motional emf
      iii. Induced E-fields
      iv. Applications of induction
   (b) AC circuits - **Chapters 14.1-14.6, 15.1-15.3**
      i. Mutual and self-inductance
      ii. Energy considerations
      iii. RL circuits
      iv. RLC circuits
7. **Unit 5:** AC Circuits and Electromagnetic Waves
   (a) AC circuits - **Chapters 15.1-15.6**
      i. AC sources and circuits
      ii. RLC series circuits
      iii. Power in AC circuits
   (b) Electromagnetic waves - **Chapters 16.1-16.3**
      i. Maxwell's equations
      ii. Electromagnetic plane waves
      iii. Energy and momentum considerations
8. **Second midterm** - due May 4th, 2026.
   (a) The midterm will be posted on Moodle on Friday, May 1st.
   (b) The exam will be completed at home, with an open-book/open-note policy. The format includes multiple choice, written exercises, and design problems.
   (c) The exam is due in PDF form on Monday, May 4th, to be submitted via Moodle.

Figure 1: (Top left) Benjamin Franklin, (Top right) Father José Antonio Alzate y Ramírez. (Bottom left) Jean-Baptiste Chappé d'Auteroche, (Bottomi right) Joaquín Velázquez de León.

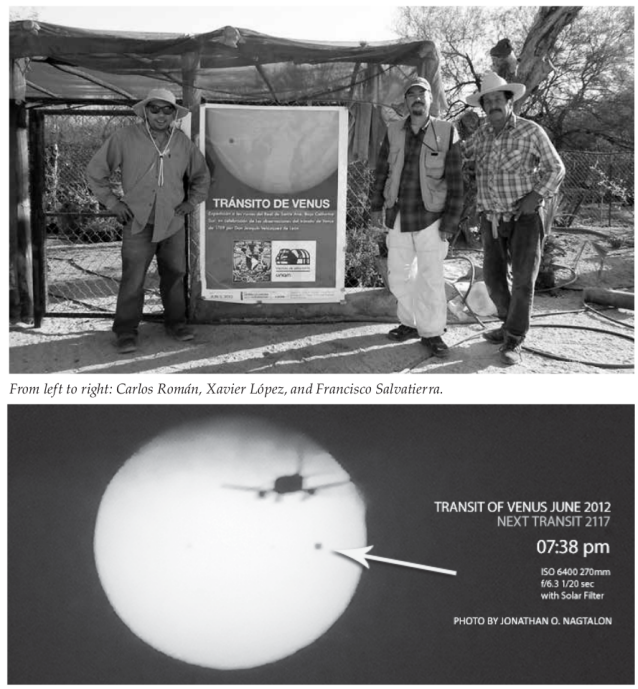

Figure 2: (Top) Carlos Román, Xavier López, and Fransisco Salvatierra. (Bottom) The 2012 Venus transit, from Baja California.

9. **Final project presentations**
   (a) Given on April 28th and 30th, teams of 1-3
   (b) Project is a self-designed physics experiment
   (c) **Option A**: A 10-15 minute traditional presentation with several minutes for questions.
   (d) **Option B**: A video in digital storytelling format using WeVideo, also 10-15 minutes long.
   (e) Students will receive training and access for the WeVideo tools, which Whittier College provides for free.
   (f) Presentations will be graded according to the following breakdown:
      i. For presenting results emphasizing *precision and attention to detail*, with *experimentally verifiable* content, 30%.
      ii. For *clearly communicating* the content, including legibility and sensible graphics, 30%.
      iii. For *meeting but not exceeding* the time-requirement with appropriate amount of content, 30%.
      iv. For *articulating the idea with a unique style*, 10%.